

Gripper control

The RealMan robotic arm supports various common grippers. In the system, the INSPIRE-ROBOTS EG2-4C2 gripper is equipped. To facilitate user operation of the gripper, the robotic arm controller has adapted the gripper for ROS-based control.

Set the force-controlled grasping of the gripper `Gripperpick.msg`

Parameter	Type	Description
<code>speed</code>	<code>uint16</code>	1–1,000, representing the gripper opening and closing speed, without a unit of measurement.
<code>force</code>	<code>uint16</code>	1–1,000, representing the gripping force of the gripper, with a maximum of 1.5 kg.
<code>block</code>	<code>uint16</code>	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.
<code>timeout</code>	<code>uint16</code>	The timeout setting for blocking mode, unit: seconds.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_gripper_pick_cmd rm_ros_interfaces/msg/Gripperpick
```

Return command demo:

```
ros2 topic echo /rm_driver/set_gripper_pick_result
```

Parameter description:

Parameter	Type	Description
<code>std_msgs</code>	<code>bool</code>	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the continuous force-controlled grasping of the gripper `Gripperpick.msg`

Parameter	Type	Description
speed	uint16	1–1,000, representing the gripper opening and closing speed, without a unit of measurement.
force	uint16	1–1,000, representing the gripping force of the gripper, with a maximum of 1.5 kg.
block	bool	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.
timeout	uint16	The timeout setting for blocking mode, unit: seconds.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_gripper_pick_on_cmd rm_ros_interfaces/msg/GripperpickOnCmd
```

Return command demo:

```
ros2 topic echo /rm_driver/set_gripper_pick_on_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the gripper to a given position Gripperset.msg

Parameter	Type	Description
position	uint16	Gripper target position, range: 1–1,000, representing the gripper opening: 0–70 mm.
block	bool	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.
timeout	uint16	The timeout setting for blocking mode, unit: seconds.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_gripper_position_cmd rm_ros_interfaces/msg/Gripperset
```

Return command demo:

```
ros2 topic echo /rm_driver/set_gripper_position_result
```



Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Control of the five-fingered dexterous hand

The RealMan RM65 robotic arm is equipped with a five-fingered dexterous hand, which can be configured by the ROS.

Set the gesture sequence number of the dexterous hand Handposture.msg

Parameter	Type	Description
posture_num	uint16	Gesture sequence number pre-stored in the dexterous hand, range: 1–40.
block	bool	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.
timeout	uint16	The timeout setting for blocking mode, unit: seconds.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_hand_posture_cmd rm_ros_interfaces/msg/Handposture
```

Return command demo:

```
ros2 topic echo /rm_driver/set_hand_posture_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the action sequence number of the dexterous hand Handseq.msg

Parameter	Type	Description
seq_num	uint16	Action sequence number pre-stored in the dexterous hand, range: 1–40.
block	bool	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.

Parameter	Type	Description
timeout	uint16	The timeout setting for blocking mode, unit: seconds.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_hand_seq_cmd rm_ros_interfaces/msg/Handseq "{seq_nu
```

Return command demo:

```
ros2 topic echo /rm_driver/set_hand_seq_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the angles of the dexterous hand Handangle.msg

Parameter	Type	Description
hand_angle	int16[6]	Array of finger angles, range: 0–1,000. Additionally, -1 indicates that no action will be performed for this degree of freedom, and the current state will be maintained.
block	bool	Blocking/Non-blocking mode, true: blocking mode, false: non-blocking mode.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_hand_angle_cmd rm_ros_interfaces/msg/Handangle "{ha
```

Return command demo:

```
ros2 topic echo /rm_driver/set_hand_angle_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the speed of the dexterous hand Handspeed.msg

Parameter	Type	Description
hand_speed	int16[6]	Finger speed, range: 1–1,000.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_hand_speed_cmd rm_ros_interfaces/msg/Handspeed "hand"
```

Return command demo:

```
ros2 topic echo /rm_driver/set_hand_speed_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.

Set the force threshold of the dexterous hand Handforce.msg

Parameter	Type	Description
hand_force	int16	Finger force, range: 1–1,000.

Usage command demo:

```
ros2 topic pub --once /rm_driver/set_hand_force_cmd rm_ros_interfaces/msg/Handforce "hand"
```

Return command demo:

```
ros2 topic echo /rm_driver/set_hand_force_result
```

Parameter description:

Parameter	Type	Description
std_msgs	bool	true: setting succeeded, false: setting failed. Return the error code from the driver terminal.